

VINAYAK GUPTA

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Professional Summary

Versatile and systems-oriented Software Engineer with 3 years of progressive experience engineering low-latency backend architectures, distributed computing infrastructure, and cross-platform native ecosystems using Go, Node.js (TypeScript), and high-concurrency PostgreSQL. Demanded at the intersection of deep computing and practical software construction, possessing a distinct multi-disciplinary background spanning runtime performance tuning, deterministic discrete mathematical algorithms, structured compiler design, and applied computational linguistics. Proven track record designing resilient cryptographic microservices, hardware-software control loops down to the 8ms execution window, and domain-specific serialization formats that radically eliminate payload network overhead. Adept at navigating legacy API specifications to map multi-layer data tokenization architectures, building multi-threaded database migration engines, and structuring declarative cross-platform mobile/desktop runtimes for applications supporting upwards of 40,000 active users.

Skills & Competencies

Core Systems Languages	Go (Golang), TypeScript, JavaScript, Python, SQL (PL/pgSQL), Java, C++, Rust, OCaml.
Backend & Architecture	Node.js (Express/TS), Django, Echo (Go), RESTful API Design, Microservices Architecture, Event-Driven Automation, Cron Pipelines, Secure Key Management Systems.
Data Layer & Cloud	PostgreSQL (High-Concurrency Optimization, Recursive CTEs, Tree Traversal Models), MongoDB, SQLite, AWS (EC2, S3), Docker, Containerized Environments (Docker Compose), Git Lifecycle.
Client & Hardware Systems	Flutter Engine, Tauri Platform (Native Linux Desktop Application Compilation), React, Embedded Hardware Orchestration (Modbus TCP, Low-Latency PLC Digital Signaling Automation Loops).
Theoretical Computing	Formal EBNF Grammar Design, Lexical Scanners & Hand-Written Recursive-Descent Parsers, Serialization Engine Architecture, Go Reflection API, Data Structure Design.
Mathematics & Analytics	Discrete Mathematical Modeling, Matrix Coordinate Transforms (Pixel-to-Millimeter Spatial Conversions), Sparse-Allocation Bitwise Algorithms, Time-Complexity Optimization ($O(1)$ Scheduling Matrices).
Linguistics & Communication	Natural Language Synthesis, Applied Computational Text Parsing, Multilingual Layout Configuration, Technical System Documentation, Interdisciplinary Cross-Team Project Management, Technical Translation.

Detailed Professional Experience

Techasoft Pvt. Ltd. <i>Systems Engineer (Field Deployment)</i>	Bangalore, India Mar 2026 – Present
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- Designed the overall software architecture for an industrial automation system, developing a Django-based backend and a native multithreaded Tauri client compiled for bare-metal Linux environments. The system was designed to operate directly alongside industrial control hardware rather than through a conventional desktop application stack.
- Developed deterministic hardware-software control loops using Python and Modbus TCP to communicate with 24V Programmable Logic Controllers (PLCs), translating software decisions into real-time physical actions within a high-speed conveyor sorting system.
- Developed a mathematical coordinate-transformation model for the computer-vision pipeline, relating image-space pixel coordinates to physical millimetre coordinates and dynamically defining Regions of Interest (ROIs). This allowed spatial information extracted from camera frames to be mapped onto the physical geometry of the manufacturing system at approximately 30 frames per second.
- Performed a production database migration from a file-based SQLite architecture to PostgreSQL, redesigning relational queries and connection management to handle concurrent writes reliably. The work required diagnosing failures arising from the interaction between application-level concurrency and database-level locking.

Medoc Health IT

Remote

Technical Lead

Feb 2025 – Jul 2025

- Led a technical team of 15–20 developers responsible for several healthcare products, coordinating system design, implementation, testing, deployment, and maintenance while establishing development processes across the full software lifecycle.
- Supported the technical onboarding of 3 multi-specialty hospitals during the first month of the engagement, helping integrate their workflows into a common healthcare platform serving more than 40,000 patients.
- Designed system architectures and interactions between independently developed services, taking responsibility for resolving technical dependencies and translating high-level product requirements into implementable engineering designs.
- Worked on the integration of government healthcare infrastructure and digital-health initiatives, including the interpretation and implementation of complex healthcare data interfaces and specifications defined by the National Health Authority (NHA).
- Worked across several substantially different layers of the technology stack, including Flutter-based mobile applications, MERN-based web and backend systems, AWS infrastructure, databases, APIs, and microservices. This required moving between application development, distributed services, infrastructure, and system integration rather than working within a single technology layer.
- Designed and maintained CI/CD and deployment workflows for microservice-based applications, taking responsibility for moving systems from development environments into production and diagnosing issues occurring across application and infrastructure boundaries.
- Contributed to the design of patient-facing healthcare applications by translating operational requirements from hospitals and healthcare users into software workflows, balancing technical constraints with usability and real-world clinical requirements.

Product Head

Feb 2023 – Jan 2025

- Worked at the intersection of software engineering, product design, and project management, leading developers while translating practical requirements from healthcare users into technical systems and software products.
- Designed system structures and application workflows by analysing how users interacted with healthcare processes, then translating those observations into technical requirements and implementation plans for the development team.
- Worked across frontend and backend systems during product development, gaining experience in understanding a system as a whole rather than as an isolated application or programming language.
- Coordinated development, testing, deployment, and iterative improvement of multiple software products, balancing engineering constraints, user requirements, project timelines, and the practical limitations of a developing organisation.
- Progressively assumed responsibility for technical decision-making and team coordination, providing the foundation for subsequently leading the organisation's technical team.

Appable Technologies

Remote

Freelance Full-Stack Software Developer

Jul 2023 – Jun 2024

- Re-architected legacy client applications by replacing tightly coupled Firebase infrastructures with independently operated backend services and PostgreSQL databases. The work involved reconsidering the architecture of existing systems rather than simply translating the implementation from one technology to another, reducing infrastructure costs by up to 65%.
- Investigated hierarchical data and permission structures in an enterprise Human Resource Management System and developed recursive PostgreSQL queries and Modified Preorder Tree Traversal (MPTT) representations to model arbitrarily deep organisational hierarchies and approval relationships.
- Designed an offline-first attendance system combining local biometric verification, geographic validation, cryptographic integrity checks, and transactional background processing. The system had to remain useful when connectivity was unavailable while preserving consistency when synchronising with the central system.
- Developed a sports-booking system involving spatial information, time-dependent pricing, membership-based pricing rules, and asynchronous communication with users. This required modelling interactions between several independent variables rather than implementing a simple fixed-price booking workflow.

Medoc Health IT

Remote

Software Development Intern

Aug 2022 – Jan 2023

- Worked on healthcare software using Flutter, JavaScript, Node.js, and SQL, contributing to both user-facing applications and backend services. This provided early experience in understanding how data, interfaces, and application logic interact across different layers of a software system.
- Investigated application defects and implemented features and maintenance fixes, developing practical experience in reading unfamiliar codebases, tracing failures through multiple components, and modifying existing systems without disrupting dependent functionality.
- Introduced containerised development environments using Docker and Docker Compose, reducing differences between individual development environments and the environments in which applications were eventually deployed.

Projects & Computational Artifacts

Zundrath Key Management System

Technologies: Go, Echo Framework, Cryptography, PBKDF2, Mutex Design, Storage Systems

- Developed a local-first, low-overhead cryptographic microservice written entirely in Go to serve highly concurrent local applications with sub-2ms storage lookup and key-extraction capabilities.
- Implemented rigorous multi-threaded safety protections using standard `sync.RWMutex` memory configurations alongside an on-demand custom disk-indexing scanner to eliminate the need for memory-heavy external database layers.
- Designed a fault-tolerant, two-phase key rotation transaction protocol that supplies a fallback 24-hour safe-migration window, guaranteeing zero decryption blackouts or race conditions during live schema updates.

Agora.Abhivyakti (Stateless Textboard Architecture)

Technologies: Go, HTML/Template Engine, Applied Linguistics, Cryptographic Hashing, Proxy Proxies

- Built a high-performance, completely stateless live-hosted web forum utilizing Go's native `html/template` compilation engine to deliver fully server-side rendered (SSR) structural layouts, eliminating all state storage overhead.
- Engineered an un-spoofable tripcode identification system utilizing custom cryptographic hashing routines that condense user secret passphrases into unique 16-character public identifiers matching restricted, regex-validated alphanumeric strings ($[A-Z][a-z][0-9]$).
- Integrated an intermediate linguistic validation and proxy security layer that intercept and parses all submitted Markdown formatting, executing strict sanitization loops and mandatory outbound-redirect warning wrappers to insulate platform visitors from malicious injection.

Grounded Object Data (GOD) Format & Compilation Suite

Technologies: Go, EBNF Grammar Design, Lexical Scanners, Recursive-Descent Parsing, Reflection API

- Conceived, specified, and engineered a completely custom, schema-less data serialization format and compiler built on a strict, formal Extended Backus-Naur Form (EBNF) grammar specification, specifically optimized to eliminate token redundancy inherent in JSON payloads.
- Hand-wrote a highly optimized lexical analyzer (scanner) and recursive-descent parser in Go, introducing a novel first-class structural paradigm for multi-row data representations: a tabular syntax (`header:rows`) that completely drops repeating object-key declarations from multi-record streams to optimize transport pipelines.
- Implemented absolute "grounded" type-safety semantics inside the runtime unmarshalling engine via Go's native reflection API, structurally preventing null-pointer runtime exceptions by auto-initializing missing, undefined, or corrupt incoming streams to zero-values or explicit `\0` null-terminators.

Publications

Book : Poetry Collection

Title : *Life*

Author : Vinayak Gupta

Published by : Sangam Publications, 2023

Patents

Vinayak Gupta (Co-author)

Title : *System for Cooking*

Application ID : 202521070227

Indian Patent Office, Published, 2025

Research Work

Research Paper Draft

- Title : *A Novel approach to Encryption for Data Security in Mobile Applications*
- Status : Manuscript draft under preparation.
- Area : Backend systems / cryptography / Secure Systems design.

Writing & Creative Work

- Maintain technical and general writing through blogs (topics include software, systems, and analytical content).
- Write fiction (short stories and narrative pieces; selected works to be published on AO3).
- Interested in combining storytelling and technical communication for educational and conceptual clarity.

Research Interests

- Early-stage independent exploration of Autism, focusing on both scientific understanding and support frameworks.
- Interest in accessibility, cognitive diversity, and designing systems for neurodivergent users.
- Drafting a research paper (manuscript in progress).

Certifications

Cyber Security and Ethical Hacking
IGTR, Indore — 2025

Education

B.Tech in Computer Science and Engineering
LNCT University, Bhopal

2021 – 2025

Technical Interests

Backend architecture, distributed systems, applied cryptography, accessibility-focused systems, programming language design